Does the ic planner pick the best configuration on curves it has not seen?

2026-10-20 · study `research/ic_planner_validation_20261020` · protocol frozen in `05a0ebd92`, amendment 1 in `d1be0c1c8`, planner `ic-planner/1` (`a29bead23`) · six ecbench sessions, 1,536 records, all verified, audits OK · reference `rho.negation`.

Question

`ic plan` and `ic bench --auto` choose every `ic.pipeline` stage from earlier measurements: the factor base and its size, the oracle, the stop rule, the target source. On prime-order curves none of that evidence was measured on, each plan ran against the alternatives one change away from it (8 targets × 3 rounds + 1 warm-up per arm and curve). Primary (amendment 1): the registered curves `icv1-fp24-tm5099-a6f137c6`, `icv1-fp26-tm1775-7e8fb6df` and `icv1-fp28-t28201-65099091` ($\log_2 r$ 24.0, 25.3, 28.0). Supplementary: the original protocol's `prime_search` curves `find_prime_order_curve(24, 3)`, `(28, 1)` and `(30, 1)` ($\log_2 r$ 23.5, 27.7, 29.6), unregistered and therefore not named by slug.

Result

In the counted unit, every choice the planner made was the better one, on all six curves.

change from the plan	r24	r26	r28	across the §5a band
base ⌜N/2⌝	1.56 [1.35, 1.77]	1.55 [1.41, 1.69]	1.43 [1.28, 1.60]	confirmed
base 2N	1.51 [1.40, 1.64]	1.60 [1.50, 1.70]	1.57 [1.49, 1.68]	**not separated**: below 1 at w = 1
`rootfind-subtract`	171 [164, 176]	233 [226, 239]	439 [424, 450]	confirmed
full-rank stop	3.64 [3.24, 4.14]	3.84 [3.43, 4.21]	4.56 [4.10, 5.12]	confirmed
random targets	44.8 [39.0, 50.6]	49.1 [43.7, 54.0]	51.5 [43.8, 58.5]	confirmed
A/A	1.000	1.000	1.000	—

S(alternative)/S(plan) on the registered curves, paired (ecbench `compare`, cluster 95% intervals, 24 pairs each). Plans: `prime-abscissa:size=N` (N = 128, 175, 323), `mitm-prime-blocked:m=2`, `incremental-gauss`, `walk`. The supplementary curves agree: half base 1.36–1.53, double 1.49–1.60, rootfind 150–637, full rank 3.30–4.81, random targets 41.5–58.7, A/A 1.

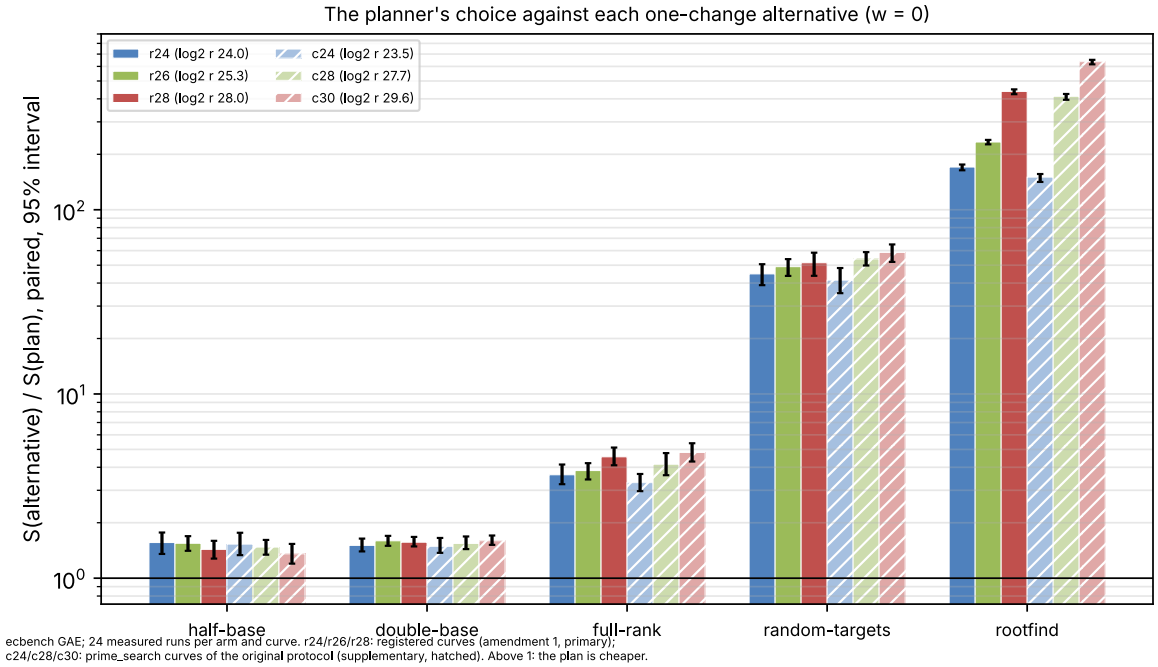


Figure 1: Each one-change alternative against the plan (w = 0), all six curves. Above 1: the plan is cheaper.

Across the pricing band the base size is the one choice that does not hold. Charging each of the five unpriced native counters (Legendre symbols, square roots, lookups, `row_ops`, guard probes) at 1 GAE, the doubled base is 2–8% cheaper than the plan on the registered curves (4–9% on the supplementary ones): a larger base needs fewer trials, and

each trial is a lookup and a guard probe. The size rule is right for the counted unit and too small if lookups cost near an addition.

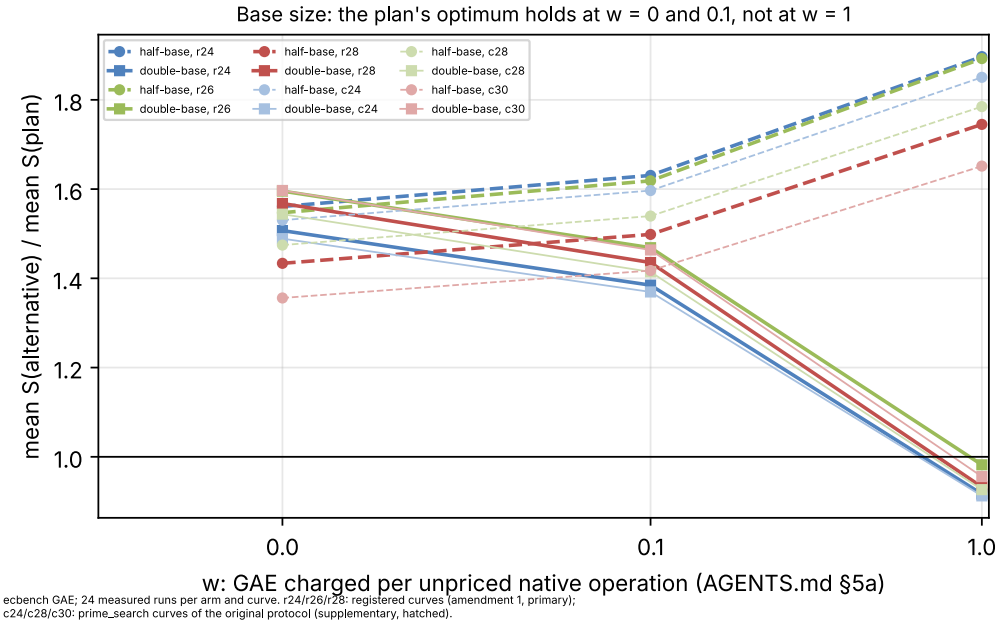


Figure 2: Half and double base against the plan across the band (ratio of mean S).

Distance remaining to rho

curve	planner's prediction	measured S(plan)/S(rho)
r24	12.2	14.4 [11.6, 18.5]
r26	14.5	14.6 [11.2, 19.6]
r28	20.1	21.7 [17.0, 28.3]
c24	11.5	12.6 [9.75, 16.5]
c28	19.3	21.8 [17.0, 29.8]
c30	24.5	25.8 [19.9, 35.2]

Each prediction lies inside the measured interval and within 18% of the point; the planner reads the gap 1–18% low. Rho is cheaper by 12.6–25.8×, as the planner's verdict said.

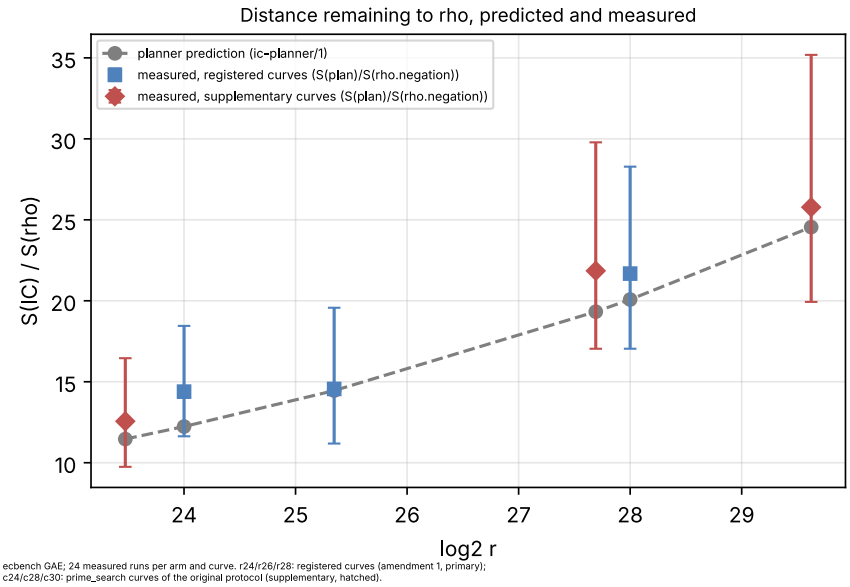


Figure 3: Predicted and measured gap to rho.

Classification

Tooling validation, engineering (AGENTS.md §3); no `S` improves on a measured baseline and no leaderboard row moves. Predictions P1–P4 held on all six curves; P5 failed for the doubled base. The oracle, stop-rule and target choices are confirmed at every band weight, and the size rule in the counted unit. Follow-up, as its own change: a weight-aware base-size rule. Deviations: amendment 1 (the original curves are unregistered; the registry on `main` is stale at 2,572 of 8,034 curves and the cover-transfer ledger pins that count); replay counts not fixed by the protocol (12 of 256 per session); a first launch on a pre-freeze binary was stopped within seconds and discarded unread.